Code No: 135BY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
COMPUTER GRAPHICS
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What are the application areas of Computer Graphics?	[2]
b)	Write a short note on work stations.	[3]
c)	What is reflection?	[2]
d)	Explain the 2D transformation matrix for Rotation.	[3]
e)	Write a short note on Hermite curve.	[2]
f)	Derive the matrix form for the scaling operation in 3-D graphics.	[3]
g)	Write about z-buffer algorithm.	[2]
h)	What are advantages of BSP tree method?	[3]
i)	What is the importance of animation in Computer Graphics?	[2]
j)	What is raster animation?	[3]

PART – B

(50 Marks)

2.a) Briefly explain about mid-point circle algorithm with example.
b) Write a short note on flood-fill algorithm. [5+5]

OR

3.a) Discuss about random scan systems.
b) Digitize the line with end points (20, 10) and (30,18) using Bresenham's algorithm. [5+5]

4.a) Describe the steps involved in Sutherland Hodgeman polygon clipping algorithm.
b) What is shear transformation? Discuss with example. [5+5]

OR

5. Derive the composite transformation matrices for translation and scaling. [10]

6.a) Write a short note on B-Spline curve.
b) Discuss about Polygon surfaces. [5+5]

OR

7.a) Briefly explain about 3D reflection.
b) Briefly explain about view volume. [5+5]

8. Discuss about basic illumination models. [10]

OR

9.a) Write a short note on area sub-division methods.
b) Explain about back-face detection. [5+5]

10.a) Explain briefly about the steps in design of animation sequence.
b) Explain about Key frame systems. [5+5]

OR

11.a) Discuss about the general purpose languages used for animation.
b) Explain about motion specifications in computer animation. [5+5]

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